Pypomp: Inference for partially observed Markov process models in Python with JAX

Aaron J. Abkemeier[1] AARONABK@UMICH.EDU
Jun Chen[2] CHENJUNC@UMICH.EDU
Kevin Tan[3] KEVTAN@WHARTON.UPENN.EDU
Jesse Wheeler[4] JESSE.WHEELER@USU.EDU
Bo Yang[5] YBB@UMICH.EDU
Kunyang He[1] KUNYANGHE@UMICH.EDU
Jonathan Terhorst[1] JONTH@UMICH.EDU
Aaron A. King[6] KINGAA@UMICH.EDU
Giles Hooker[3] GHOOKER@WHARTON.UPENN.EDU
Edward L. Ionides[1] IONIDES@UMICH.EDU

[1] *University of Michigan, Department of Statistics;* [2] *Duke University, Department of Statistics;* [3] *University of Pennsylvania, Department of Statistics;* [4] *Utah State University, Department of Mathematics & Statistics;* [5] *Columbia University, Department of Marketing;* [6] *University of Michigan, Department of Ecology & Evolutionary Biology*

Editor:

Abstract

Nonlinear, non-Gaussian partially observed Markov process (POMP) models arise in various areas including epidemiology, ecology, and robotics. Model-based inferences typically involve computationally intensive Monte Carlo methods. We introduce Pypomp, a Python library for data analysis via POMP models that uses JAX to take advantage of GPU hardware, automatic differentiation, and just-in-time compilation. By design, Pypomp supports model development and data analysis as well as method development, following the approach of the pomp R package. However, Pypomp obtains considerably higher performance: Pypomp running on one GPU can filter as many particles per second as 16,000 CPU cores running pomp. A central algorithm for Pypomp is the particle filter, and Pypomp is faster on our benchmark tests than other GPU-compatible particle filter libraries. The particle filter in Pypomp is differentiable, with a tunable bias-variance tradeoff.

Keywords: pomp, JAX, GPU acceleration, likelihood-based inference

1 Introduction

Partially observed Markov process (POMP) models provide a flexible framework for modeling dynamic systems whose underlying state is not directly observed. POMP models, also known as state-space models or hidden Markov models, consist of a latent Markov process and incomplete or noisy observations. The task of evaluating the likelihood function and estimating latent states for a POMP model is called the filtering problem.

Pypomp is a Python library combining advances in hardware, software, and algorithms for POMP models, with an emphasis on inference via the particle filter (PF). PF, also

known as sequential Monte Carlo, provides an estimate of the likelihood of a POMP model (Arulampalam et al., 2002). The basic PF algorithm needs to simulate from the process model but does not evaluate its transition probabilities, a property known as plug-and-play (P&P) (Bretó et al., 2009). Pypomp focuses on P&P methods because they facilitate scientific modeling, enabling applications in epidemiology (Mietchen et al., 2024; Fox et al., 2022; Wen et al., 2024), ecology (Auger-Méthé et al., 2021; Marino et al., 2019; Blackwood et al., 2013), finance (Bretó, 2014), robotics (Jonschkowski et al., 2018) and other domains where nonlinear mechanistic stochastic dynamic models are pertinent. Pypomp provides P&P inference methods extending PF: maximum likelihood parameter estimation by iterated filtering (IF2) (Ionides et al., 2015), Bayesian inference by particle Markov chain Monte Carlo (Andrieu et al., 2010), and stochastic gradient descent (SGD) via a differentiable particle filter (DPF) algorithm called a measurement-off-parameter (MOP) particle filter (Tan et al., 2026).

Pypomp provides a framework for specifying models and inference methods, a range of pre-built models and methods, and various graphical and diagnostic tools to assist practical data analysis. Thus, Pypomp supports complete data analysis workflows, facilitating the evaluation of a range of models and statistical methods. Models and methods share an interface so that new models can readily be tested on pre-existing methods, and new methods can be tested on pre-existing models. These design features are inherited from the R package pomp (King et al., 2016). Although the method interface of Pypomp (and pomp) is not limited to P&P algorithms, experience has shown that P&P approaches are convenient for scientific data analysis. Theoretical obstacles have limited the practical scope of non-P&P algorithms (Roberts and Stramer, 2001; Karppinen et al., 2024).

Pypomp leverages JAX (Bradbury et al., 2018) to provide just-in-time (JIT) compilation, automatic differentiation (AD), and the capability to run on massively parallel hardware such as GPUs. When the state space is small and discrete, or sufficiently close to linear Gaussian for applicability of the extended or unscented Kalman filter, the Dynamax library (Linderman et al., 2025) provides JAX-based solutions. Other JAX-based libraries providing PF and DPF methods for POMP models include PFJAX (Lysy et al., 2022) and Cuthbert (Duffield, 2026). PyDPF (Brady et al., 2026) provides a DPF using PyTorch and focuses on POMP models parameterized using neural networks. The NumPy-based libraries particles (Chopin and Papaspiliopoulos, 2020), a general sequential Monte Carlo framework, and pypfilt (Moss, 2024) provide PF without AD. Pypomp distinguishes itself from these other packages by (i) following the design decisions of pomp that support a full data analysis workflow; (ii) superior performance on benchmark tests. Pypomp is not limited to PF methods, and includes, for example, the approximate Bayesian computing (ABC) P&P algorithm of Marjoram et al. (2003). However, PF has proved an effective tool for the development and analysis of POMP models in scientific contexts.

Pypomp compiles models with JAX's XLA compiler by default; on NVIDIA GPUs, an optional CuTe backend translates them into CUDA kernels (Thakkar et al., 2023; Cecka, 2026). With enough particles and method replications, one GPU running the CuTe backend runs iterated filtering as fast as 9,744 to 16,120 CPU cores running pomp (Appendix C). Such replications are routine in practice: practical likelihood maximization involves searching from many initial parameter vectors, and Monte Carlo adjusted profile (MCAP) confidence intervals (Ionides et al., 2017), implemented in Pypomp as `mcap`, require maxi-

mizations. Profile likelihood approaches have been found preferable to Fisher information for confidence interval construction for linear Gaussian POMP models (Wheeler and Ionides, 2025). MCAP algorithms can lead to efficient inference even in a limit where the Monte Carlo error grows with the amount of data (Ning et al., 2021).

Algorithms to take advantage of JIT, AD, and GPU are an area of active development. Pypomp implements the MOP P&P DPF algorithm of Tan et al. (2026), with the resulting derivative being called DMOP implemented within the `train` method. DMOP is competitive on benchmark problems when combined with Adam SGD, especially when using IF2 for a warm start (Tan et al., 2026). Pypomp provides inverse transform sampling methods for use in `rproc` and `rmeas` since the rejection sampling used in some `jax.random` samplers is inefficient on GPUs (Giles, 2016; Giles and Beentjes, 2024; Temme, 1992). These samplers run 23 to 45 times faster than `jax.random` on a GPU. Quantiles of inverse and rejection sampling draws differ by at most 1.1% of a standard deviation on average (Appendix D).

2 POMP Modeling in Pypomp

Let $\{X(t), t_0 \le t \le t_N\}$ denote the latent process, and write $Y_{1:N}$ for the measurement process at times $t_{1:N}$. We suppose Y_n is conditionally independent of the other latent states and measurements given $X_n = X(t_n)$. The data are observed values, $y^*_{1:N}$. A POMP model is specified by its initial state distribution, $f_{X_0}(x_0;\theta)$, state transition density $f_{X_{n+1}|X_n}(x_{n+1}|x_n;\theta)$, and measurement density $f_{Y_n|X_n}(y_n|x_n;\theta)$, where θ denotes a vector of model parameters. For P&P methods, it suffices to specify a simulator for $f_{X_0}(x_0;\theta)$ known as `rinit`, a simulator for $f_{X_{n+1}|X_n}(x_{n+1}|x_n;\theta)$ known as `rproc`, a simulator for $f_{Y_n|X_n}(y_n|x_n;\theta)$ known as `rmeas`, and an evaluator for $f_{Y_n|X_n}(y_n|x_n;\theta)$ known as `dmeas`. The user passes JAX-compatible code for these functions, together with the data, to build a `Pomp` class model object using the `Pomp` constructor function. Tutorials are available at `https://github.com/pypomp/tutorials`. Pypomp includes models for historical monthly cholera mortality in Dhaka, Bangladesh (`pypomp.models.dacca`), historical weekly measles cases in London, UK (`pypomp.models.measles.pomp`), daily log returns of the S&P 500 index (`pypomp.models.spx`), and a panel model for measles cases in up to 1,422 urban and rural districts in England and Wales (`pypomp.models.measles.panel_pomp`).

A collection of POMP models is a PanelPOMP (Bretó et al., 2020), and each constituent POMP is called a *unit*. Parameters may be unit-specific or shared between units. Marginalized panel iterated filtering (MPIF) (Wheeler et al., 2025) and DMOP provide likelihood maximization for models of class `PanelPomp`.

Below, we demonstrate maximum likelihood estimation with IF2 (`mif`) followed by refinement using SGD (`train`) and then log-likelihood evaluation using PF (`pfilter`), on the `spx` model. Each method stores its convergence records, execution time, and arguments in the `results_history` attribute of the `Pomp` object, enabling sequential method chaining and diagnostic investigation by helper methods such as `traces`.

```
import pypomp as pp, jax
spx_obj = pp.models.spx()
spx_obj.mif(J=1000, M=10, key=jax.random.key(1), rw_sd=pp.RWSigma(
    sigmas={"mu": 0.02, "kappa": 0.02, "theta": 0.02,
```

Table 1: Wall-clock time in seconds for 36 IF2 searches, with pomp on 36 CPU cores and Pypomp 1.4.0 on one RTX PRO 6000 Blackwell GPU. Cores, the CPU cores running pomp that the GPU matched, is 36 times the pomp time divided by the Pypomp time; Peak is its highest value in scaling tests (Appendix C).

	pomp	Pypomp, XLA (1 GPU)			Pypomp, CuTe (1 GPU)		
Model	(36 cores)	Time	Cores	Peak	Time	Cores	Peak
Cholera, Dhaka	1,609	28.5	2,035	3,128	37.9	1,530	16,120
Measles, London	2,923	31.8	3,310	5,046	68.6	1,534	12,711
Measles, 4-town panel	8,262	97.6	3,047	3,697	105	2,836	9,744
S&P 500	643	35.0	662	8,360	25.4	913	12,297

```
            "xi": 0.02, "rho": 0.02, "V_0": 0.1},
    init_names=["V_0"]).geometric_cooling(a=0.5))
spx_obj.train(J=1000, M=10, optimizer=pp.Adam(), eta=pp.LearningRate(
    {name: 0.001 for name in spx_obj.canonical_param_names}
    ).geometric_decay(decay_rate=0.5, M=10))
spx_obj.pfilter(J=1000, reps=10)
```

Two performance questions are: (i) what is the benefit of using GPU hardware versus CPU? (ii) how does Pypomp compare in speed to alternative GPU-based particle filter libraries? To address (i), we compare Pypomp on a GPU with pomp on a CPU because that informs a natural decision for scientists considering Pypomp as a data analysis tool. With model code in compiled C snippets, pomp has comparable speed to other high-performance CPU-based packages for POMP models (FitzJohn et al., 2020). Table 1 shows four models with different characteristics (details in Appendix A and `https://github.com/pypomp/quant`). Efficiency peaks when the GPU is saturated, and for small tasks CuTe's longer compilation can outweigh its faster iterations (Appendix C). With `jax.random` samplers instead of `pypomp.random`, IF2 for the London measles model took 45 times longer (Appendix D). To address (ii), Appendix B runs the same particle filter workloads in five other Python libraries; on the London measles model, once compiled, `pfjax` and `cuthbert` took about 84 times as long as Pypomp.

3 Governance and Acknowledgments

The source code for Pypomp is openly available at `https://github.com/pypomp/pypomp`. The project is run by the Pypomp organization (`https://github.com/pypomp`) under democratic governance rules that welcome new developers and community contributions. The Pypomp project is supported by National Science Foundation grants OAC-2607869 and OAC-2607870. The developers acknowledge coding assistance from artificial intelligence models by OpenAI, Anthropic and Google.

References

Christophe Andrieu, Arnaud Doucet, and Roman Holenstein. Particle Markov chain Monte Carlo methods. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 72(3):269–342, 2010. doi: 10.1111/j.1467-9868.2009.00736.x.

M. Sanjeev Arulampalam, Simon Maskell, Neil Gordon, and Tim Clapp. A tutorial on particle filters for online nonlinear/non-Gaussian Bayesian tracking. *IEEE Transactions on Signal Processing*, 50(2):174–188, 2002. doi: 10.1109/78.978374.

Marie Auger-Méthé, Ken Newman, Diana Cole, Fanny Empacher, Rowenna Gryba, Aaron A. King, Vianey Leos-Barajas, Joanna Mills Flemming, Anders Nielsen, Giovanni Petris, and Len Thomas. A guide to state-space modeling of ecological time series. *Ecological Monographs*, 91(4):e01470, 2021. doi: 10.1002/ecm.1470.

Alexander D. Becker and Bryan T. Grenfell. tsiR: An R package for time-series Susceptible-Infected-Recovered models of epidemics. *PLOS ONE*, 12(9):e0185528, 2017. doi: 10.1371/journal.pone.0185528.

Julie C. Blackwood, Daniel G. Streicker, Sonia Altizer, and Pejman Rohani. Resolving the roles of immunity, pathogenesis, and immigration for rabies persistence in vampire bats. *Proceedings of the National Academy of Sciences of the United States of America*, 110 (51):20837–20842, 2013. doi: 10.1073/pnas.1308817110.

James Bradbury, Roy Frostig, Peter Hawkins, Matthew James Johnson, Chris Leary, Dougal Maclaurin, George Necula, Adam Paszke, Jake VanderPlas, Skye Wanderman-Milne, and Qiao Zhang. JAX: Composable transformations of Python+NumPy programs, 2018.

John-Joseph Brady, Benjamin Cox, Yunpeng Li, and Víctor Elvira. PyDPF: A Python package for differentiable particle filtering. *arXiv:2510.25693*, 2026. doi: 10.48550/arXiv.2510.25693.

Carles Bretó. On idiosyncratic stochasticity of financial leverage effects. *Statistics & Probability Letters*, 91:20–26, 2014. doi: 10.1016/j.spl.2014.04.003.

Carles Bretó, Daihai He, Edward L. Ionides, and Aaron A. King. Time series analysis via mechanistic models. *The Annals of Applied Statistics*, 3(1):319–348, 2009. doi: 10.1214/08-AOAS201.

Carles Bretó, Edward L. Ionides, and Aaron A. King. Panel Data Analysis via Mechanistic Models. *Journal of the American Statistical Association*, 115(531):1178–1188, 2020. doi: 10.1080/01621459.2019.1604367.

Cris Cecka. CuTe layout representation and algebra. *arXiv:2603.02298*, 2026. doi: 10.48550/arXiv.2603.02298.

Nicolas Chopin and Omiros Papaspiliopoulos. *An Introduction to Sequential Monte Carlo*. Springer, 2020. doi: 10.1007/978-3-030-47845-2.

Sam Duffield. cuthbert: State-space model inference with JAX, 2026. URL `https://pypi.org/project/cuthbert/`.

Richard G. FitzJohn, Edward S. Knock, Lilith K. Whittles, Pablo N. Perez-Guzman, Sangeeta Bhatia, Fernando Guntoro, Oliver J. Watson, Charles Whittaker, Neil M. Ferguson, Anne Cori, Marc Baguelin, and John A. Lees. Reproducible parallel inference and simulation of stochastic state space models using odin, dust, and mcstate. *Wellcome Open Research*, 5, 2020. doi: 10.12688/wellcomeopenres.16466.2.

Spencer J. Fox, Michael Lachmann, Mauricio Tec, Remy Pasco, Spencer Woody, Zhanwei Du, Xutong Wang, Tanyi A. Ingle, E. Javan, Maytal Dahan, Kelly Gaither, Mark E. Escott, Stephen I. Adler, S. Claiborne Johnston, James G. Scott, and Lauren A. Meyers. Real-time pandemic surveillance using hospital admissions and mobility data. *Proceedings of the National Academy of Sciences*, 119(7):e2111870119, 2022. doi: 10.1073/pnas.2111870119.

Michael B. Giles. Algorithm 955: Approximation of the Inverse Poisson Cumulative Distribution Function. *ACM Transactions on Mathematical Software*, 42(1):1–22, 2016. doi: 10.1145/2699466.

Michael B. Giles and Casper Beentjes. Approximation of an inverse of the incomplete Beta Function. In Kevin Buzzard, Alicia Dickenstein, Bettina Eick, Anton Leykin, and Yue Ren, editors, *Mathematical Software – ICMS 2024*, volume 14749 of *Lecture Notes in Computer Science*, pages 207–214. Springer Nature Switzerland, Cham, 2024. doi: 10.1007/978-3-031-64529-7_22.

Daihai He, Edward L. Ionides, and Aaron A. King. Plug-and-play inference for disease dynamics: Measles in large and small populations as a case study. *Journal of The Royal Society Interface*, 7(43):271–283, 2010. doi: 10.1098/rsif.2009.0151.

Edward L. Ionides, Dao Nguyen, Yves Atchadé, Stilian Stoev, and Aaron A. King. Inference for dynamic and latent variable models via iterated, perturbed Bayes maps. *Proceedings of the National Academy of Sciences*, 112(3):719–724, 2015. doi: 10.1073/pnas.1410597112.

Edward L. Ionides, Carles Breto, Joonha Park, R. Alexander Smith, and Aaron A. King. Monte Carlo profile confidence intervals for dynamic systems. *Journal of The Royal Society Interface*, 14(132):20170126, 2017. doi: 10.1098/rsif.2017.0126.

Rico Jonschkowski, Divyam Rastogi, and Oliver Brock. Differentiable particle filters: End-to-end learning with algorithmic priors. *Robotics: Science and Systems XIV*, 2018. doi: 10.48550/arXiv.1805.11122.

Santeri Karppinen, Sumeetpal S Singh, and Matti Vihola. Conditional particle filters with bridge backward sampling. *Journal of Computational and Graphical Statistics*, 33(2):364–378, 2024. doi: 10.1080/10618600.2023.2231514.

Aaron A. King, Edward L. Ionides, Mercedes Pascual, and Menno J. Bouma. Inapparent infections and cholera dynamics. *Nature*, 454(7206):877–880, 2008. doi: 10.1038/nature07084.

Aaron A. King, Dao Nguyen, and Edward L. Ionides. Statistical Inference for Partially Observed Markov Processes via the *R* Package **pomp**. *Journal of Statistical Software*, 69(12), 2016. doi: 10.18637/jss.v069.i12.

Scott W. Linderman, Peter Chang, Giles Harper-Donnelly, Aleyna Kara, Xinglong Li, Gerardo Duran-Martin, and Kevin Murphy. Dynamax: A python package for probabilistic state space modeling with JAX. *Journal of Open Source Software*, 10(108):7069, 2025. doi: 10.21105/joss.07069.

Martin Lysy, Pranav Subramani, Jonathan Ramkissoon, Mohan Wu, Michelle Ko, and Kanika Choptra. PFJAX: Particle filtering in JAX, 2022. URL `https://github.com/mlysy/pfjax`.

John A. Marino, Jr., Scott D. Peacor, David B. Bunnell, Henry A. Vanderploeg, Steven A. Pothoven, Ashley K. Elgin, James R. Bence, Jing Jiao, and Edward L. Ionides. Evaluating consumptive and nonconsumptive predator effects on prey density using field time-series data. *Ecology*, 100(3):e02583, 2019. doi: 10.1002/ecy.2583.

Paul Marjoram, John Molitor, Vincent Plagnol, and Simon Tavaré. Markov chain Monte Carlo without likelihoods. *Proceedings of the National Academy of Sciences*, 100(26): 15324–15328, 2003. doi: 10.1073/pnas.0306899100.

Matthew S. Mietchen, Erin Clancey, Corrin McMichael, and Eric T. Lofgren. Estimating SARS-CoV-2 transmission parameters between coinciding outbreaks in a university population and the surrounding community. medRxiv preprint doi: 10.1101/2024.01.10.24301116, 2024.

Robert Moss. pypfilt: A particle filter for Python. *Journal of Open Source Software*, 9(96): 6276, 2024. doi: 10.21105/joss.06276.

N. Ning, E. L. Ionides, and Y. Ritov. Scalable Monte Carlo inference and rescaled local asymptotic normality. *Bernoulli*, 27:2532–2555, 2021. doi: 10.3150/20-BEJ1321.

G. O. Roberts and O. Stramer. On inference for partially observed nonlinear diffusion models using the Metropolis-Hastings algorithm. *Biometrika*, 88:603–621, 2001. doi: 10.1093/biomet/88.3.603.

John K. Salmon, Mark A. Moraes, Ron O. Dror, and David E. Shaw. Parallel random numbers: As easy as 1, 2, 3. In *Proceedings of 2011 International Conference for High Performance Computing, Networking, Storage and Analysis*, pages 16:1–16:12. Association for Computing Machinery, 2011. doi: 10.1145/2063384.2063405.

Kevin Tan, Aaron J. Abkemeier, Kunyang He, Giles Hooker, and Edward L. Ionides. Automatic differentiation for inference on partially observed Markov process models. *arXiv:2407.03085v2*, 2026. doi: 10.48550/arXiv.2407.03085.

Nico M. Temme. Asymptotic inversion of incomplete gamma functions. *Mathematics of Computation*, 58(198):755–764, 1992. doi: 10.1090/S0025-5718-1992-1122079-8.

Vijay Thakkar, Pradeep Ramani, Cris Cecka, Aniket Shivam, Honghao Lu, Ethan Yan, Jack Kosaian, Mark Hoemmen, Haicheng Wu, Andrew Kerr, Matt Nicely, Duane Merrill, Dustyn Blasig, Aditya Atluri, Fengqi Qiao, Piotr Majcher, Paul Springer, Markus Hohnerbach, Jin Wang, and Manish Gupta. CUTLASS, January 2023. URL `https://github.com/NVIDIA/cutlass`.

Li Wen, Yi Yin, Qiong Li, Zhihang Peng, and Daihai He. Modeling the co-circulation of influenza and COVID-19 in Hong Kong, China. *Advances in Continuous and Discrete Models*, 2024:32, 2024. doi: 10.1186/s13662-024-03830-7.

Jesse Wheeler and Edward L Ionides. Revisiting inference for ARMA models: Improved fits and superior confidence intervals. *PloS One*, 20(10):e0333993, 2025. doi: 10.1371/journal.pone.0333993.

Jesse Wheeler, Aaron J. Abkemeier, and Edward L. Ionides. Iterating marginalized Bayes maps for likelihood maximization with application to nonlinear panel models. *arXiv:2511.17438*, 2025. doi: 10.48550/arXiv.2511.17438.

Appendix A. Full benchmark results

The four benchmark models stress different parts of the particle filter, which helps locate performance bottlenecks.

Cholera, Dhaka. The cholera model of King et al. (2008) has 600 monthly observations with 20 Euler steps between each, and each step updates a stochastic differential equation with one normal draw. The process model is cheap, but there are many steps between resamples, so this model isolates the per-step overhead inside the process model.

Measles, London. The measles model of He et al. (2010) has weekly case reports from 1950 to 1963 and takes a daily Euler step, interpolating covariates and drawing binomial, gamma, and Poisson random variables at every step. It has a relatively expensive process model, so it isolates how the GPU handles heavy computation per particle, while testing the fast samplers in `pypomp.random` (Appendix D).

Measles, 4-town panel. The panel model fits the same measles model to London, Halesworth, Hastings, and Cardiff jointly, with some parameters shared across towns and the rest specific to each town. MPIF (Wheeler et al., 2025) filters the towns sequentially within each iteration, so an iteration costs about four single-town iterations with the same number of particles in flight. This model tests the panel methods used in Appendix E. With the default XLA backend, its IF2 ratio of R pomp CPU time to GPU time is lower than for the London model because R runs it relatively fast, likely because the random variate generators in R can take a cheaper route for the smaller parameter values of the smaller towns.

S&P 500. The stochastic volatility model of Bretó (2014) has 3,523 daily log returns and takes one step of its process model per observation, with a single normal draw. Each step between resamples is very cheap, so fixed per-observation costs such as resampling can dominate.

Table 1 benchmarks Pypomp 1.4.0 against pomp 6.3 and panelPomp 1.7.0.0. R pomp and Pypomp run the same workload in a given benchmark. CPU runs used 36-core nodes

Table 2: Wall-clock time in seconds for IF2 and for particle filter (PF) evaluation, with the workloads described above, and with Pypomp 1.4.0. Pypomp IF2 and PF times include JIT compilation. Pypomp GPU times are with its default XLA backend and with its CuTe backend. Software ratio is the pomp (or panelPomp) time divided by the Pypomp time on a 36-core node, and total ratio is the pomp time divided by the Pypomp GPU time with each backend. These workloads do not saturate the GPU, so the total ratios understate its throughput; the peak columns of Table 1 give the GPU's highest measured IF2 throughput. The 4-town panel has no Pypomp CPU run.

		`pomp`	Pypomp			Ratio		
				1 GPU			Total	
Model	Task	36 cores	36 cores	XLA	CuTe	Software	XLA	CuTe
Cholera, Dhaka	IF2	1,609	414	28.5	37.9	3.9×	57×	42×
	PF	380	67.7	4.1	6.5	5.6×	93×	58×
Measles, London	IF2	2,923	1,116	31.8	68.6	2.6×	92×	43×
	PF	895	308	14.7	6.1	2.9×	61×	148×
Measles, 4-town panel	IF2	8,262	—	97.6	105	—	85×	79×
	PF	2,252	—	32.7	13.3	—	69×	169×
S&P 500	IF2	643	281	35.0	25.4	2.3×	18×	25×
	PF	32.0	7.8	1.9	1.7	4.1×	17×	19×

with Intel Xeon Gold 6154 or Gold 6254 processors, and GPU runs used one NVIDIA RTX PRO 6000 Blackwell Server Edition. Each benchmark runs 36 IF2 searches from the same starting points and then runs replicate particle filters at each search's IF2 estimate, with as many particles per filter as per search. The cholera and both measles benchmarks use 5,000 particles, 100 IF2 iterations, and 36 particle filter replicates, and the S&P 500 benchmark uses 1,000 particles, 200 iterations, and 24 replicates.

Fitting the measles model of He et al. (2010) to all 1,422 towns and cities for 1950–1963 took one GPU running the CuTe backend 15 hours, including likelihood evaluation, versus an estimated 4.8 months for R on a 36-core node, extrapolating linearly from panelPomp's cost on the 4-town panel (Appendix E). Thus, Pypomp enables a scientific analysis that was previously considered infeasible (Becker and Grenfell, 2017).

Table 2 extends Table 1 with IF2 and particle filter times for Pypomp on a 36-core CPU node like those used for R pomp. Pypomp on CPU was 2.3 to 5.6 times faster than R pomp, and moving Pypomp to the GPU with the default XLA backend makes it another 4.1 to 35 times faster. These workloads are sized for R and too small to saturate the GPU, so the GPU ratios understate what the GPU can do. At these sizes, the CuTe backend's longer compilation outweighs its faster iterations: its IF2 was slower than XLA's for every model except the S&P 500. At its peak throughput in the scaling tests (Appendix C), one GPU runs IF2 87 to 232 times as fast as R on the 36-core node with the default XLA backend, and 271 to 448 times as fast with the CuTe backend.

Appendix B. Comparison with other particle filter libraries

Table 3 times the particle filter workloads of Table 2 in other Python particle filter libraries, with each model written as a user of that library would write it. Every library runs the bootstrap particle filter with the same number of filters and particles as in Table 2. The GPU libraries ran on one RTX PRO 6000 Blackwell Server Edition, the GPU model used in Table 1, with the same Pypomp release, but each library's timing script ran separately from the benchmarks in Table 2, so the Pypomp times in the two tables differ somewhat. We left out Dynamax (Linderman et al., 2025), since its only particle filter is a Rao–Blackwellized filter for switching linear Gaussian models.

Table 3 gives two times for each GPU library: a call with the filter already compiled, and the first call, which includes JIT compilation. IF2 compiles its filter once and then runs it at every iteration, so in an IF2 search with hundreds of iterations the compiled time dominates, and we compare the libraries by it below, against Pypomp with its default XLA backend.

On the cholera model, `pfjax` and `cuthbert` took 1.3 to 1.4 times as long as Pypomp. On the measles models, `pfjax` and `cuthbert` took about 84 times as long as Pypomp for London, and neither finished its first call on the 4-town panel within an hour. Most of this gap comes from the `jax.random` rejection samplers, which these libraries use for the binomial, gamma, and Poisson draws of the measles model (Appendix D). The PyTorch library `pydpf` took 5.9 to 9.2 times as long as Pypomp on the cholera and measles models. On the S&P 500 model, whose steps are cheap, every GPU library's first call took under two seconds, and `pydpf`'s first call, at 1.02 seconds, was slightly faster than Pypomp's 1.07 seconds. Once compiled, Pypomp was the fastest GPU library on all four models, and its CuTe backend ran the compiled filter a further 1.7 to 5.2 times as fast as its default XLA backend.

The CPU-only libraries `particles` and `pypfilt` were timed on one core, and we scaled each time to 36 cores assuming perfect scaling, which flatters them. Under this assumption, `particles` took 0.5 to 1.0 times the measured pomp time, and `pypfilt` took 1.0 to 9.4 times the pomp time.

Appendix C. GPU scaling tests

The benchmarks in Table 1 and Table 2 use the R configuration, which does not give the GPU enough work to reach its highest throughput for every model. The scaling tests in quant ran IF2 on one RTX PRO 6000 Blackwell Server Edition GPU with more particles in flight, added in two ways: more searches of 1,000 particles each, or more particles for each of a fixed number of searches. Each configuration, with K searches of J particles each (KJ total particles in flight), ran in its own process and ran a short IF2 search of m iterations twice, taking t_{cold} seconds on the first run and t_{warm} seconds on the second (excluding JIT compilation). Projecting to the benchmark's M IF2 iterations with compilation included gives the throughput, in particles filtered per second,

$$\frac{KJM}{(t_{\text{cold}} - t_{\text{warm}}) + M\, t_{\text{warm}}/m}.$$

Table 3: Wall-clock time in seconds for the particle filter benchmarks of Table 2 in Pypomp 1.4.0 with its default XLA backend and with its CuTe backend and Philox generator, `pfjax` 0.0.3rc1 and `cuthbert` 0.1.1 (JAX), `pydpf` 1.2.3 (PyTorch), and `particles` 0.3 and `pypfilt` 0.8.7 (NumPy). GPU times are from one RTX PRO 6000 Blackwell Server Edition; for each model, the first row is the time for a call with the filter already compiled, and the second row, in parentheses, is the first call, which includes JIT compilation. (> 3,600) means the first call did not finish within an hour, so there is no compiled time. For the measles models, `pfjax` and `cuthbert` draw with the `jax.random` samplers (Appendix D). †`particles` and `pypfilt` run only on CPUs; their times are the time for one filter on one core, multiplied by the number of filters and divided by 36. *`panelPomp` for the 4-town panel, as in Table 2.

	1 GPU					36 CPU cores		
	Pypomp							
Model	XLA	CuTe	`pfjax`	`cuthbert`	`pydpf`	`particles`†	`pypfilt`†	`pomp`*
Cholera, Dhaka	4.1	0.80	5.4	5.8	24.1	203	429	380
	(5.4)	(4.2)	(7.9)	(6.8)	(24.8)			
Measles, London	6.1	2.6	512	513	56.3	450	897	895
	(11.7)	(5.0)	(518)	(516)	(56.5)			
Measles, 4-town panel	23.4	9.3	—	—	144	2,166	4,393	2,252
	(30.0)	(12.1)	(>3,600)	(>3,600)	(144)			
S&P 500	0.27	0.16	0.59	1.2	0.89	16.8	301	32.0
	(1.1)	(1.1)	(1.4)	(1.8)	(1.0)			

Each core in the pomp benchmark ran one search of J_{R} particles for M iterations in t_{R} seconds, giving a single-core throughput of $J_{\mathrm{R}}M/t_{\mathrm{R}}$; dividing the GPU throughput by this rate gives the number of CPU cores running pomp that the GPU matches. The highest value of this ratio over the scaling tests is the XLA Peak column of Table 1, compared there and in Figure 1 against the Cores column measured directly in the full-length timing benchmark, where $K = 36$, $J = J_{\mathrm{R}}$, and t_{GPU} includes compilation over all M iterations without projection, reducing the ratio to $36\,t_{\mathrm{R}}/t_{\mathrm{GPU}}$. The scaling comparisons assume that pomp's time per particle per iteration does not depend on the number of particles or searches.

Figure 1 shows the results. With the default XLA backend, for every model, throughput peaks at between 1.0 and 2.4 million particles and falls or plateaus at larger sizes. The peak is 1.5 times the benchmark for the cholera model, 1.5 times for the London measles model, 1.2 times for the 4-town panel, and 12.6 times for the S&P 500 model. The S&P 500 model gains the most because its benchmark, with cheap steps and few particles, leaves the GPU mostly idle. For this model, adding searches beats adding particles per search at a given total number of particles by an average of 7%. Consequently, there is not a large difference in speed between adding searches or particles per search on the GPU in terms of scaling with total particles.

The results above use JAX's default XLA compiler, which runs each Euler step of the process model as a sequence of GPU kernels that pass intermediate values through GPU memory. Pypomp 1.4.0 also has an optional CuTe backend, which uses the CuTe DSL of NVIDIA's CUTLASS library (Thakkar et al., 2023; Cecka, 2026) to translate the process and measurement models into CUDA kernels that keep each particle's state on the chip between observations, and draws random numbers with Pypomp's implementation of the Philox4x32-10 generator of Salmon et al. (2011). We reran the same scaling tests with this backend, extended to as many as 38.4 million particles. Its compilation takes longer, 9 to 55 seconds per configuration against 3 to 12 seconds for XLA, but its IF2 iterations are faster, and its highest throughput matches 9,744 to 16,120 CPU cores running R, the CuTe peak column of Table 1. CuTe's throughput keeps rising to much larger sizes than XLA's (Figure 1). For the S&P 500 model, it peaks at 8.0 million particles when adding searches and 9.6 million when adding particles per search, and then falls. For the other models, it changed by less than 3% over the last doubling of particles, so larger configurations are unlikely to match many more cores.

With the CuTe backend, each million particles used about 133 MB for the S&P 500 model, 265 MB for the 4-town panel, 293 MB for the London measles model, and 640 MB for the cholera model, whether particles were added as searches or per search. At the largest configurations it ran, the default XLA backend used about as much memory as CuTe.

Appendix D. Random variable samplers

`jax.random` draws Poisson, binomial, and gamma random variables with rejection samplers, which loop a random number of times per draw. This suits GPUs poorly, since threads that run together wait for the slowest one. `pypomp.random` instead provides approximate inverse-CDF samplers, which avoid rejection loops. Table 4 compares the two on 1,000,000 draws per call, with parameters spread across the values where each sampler switches algorithms,

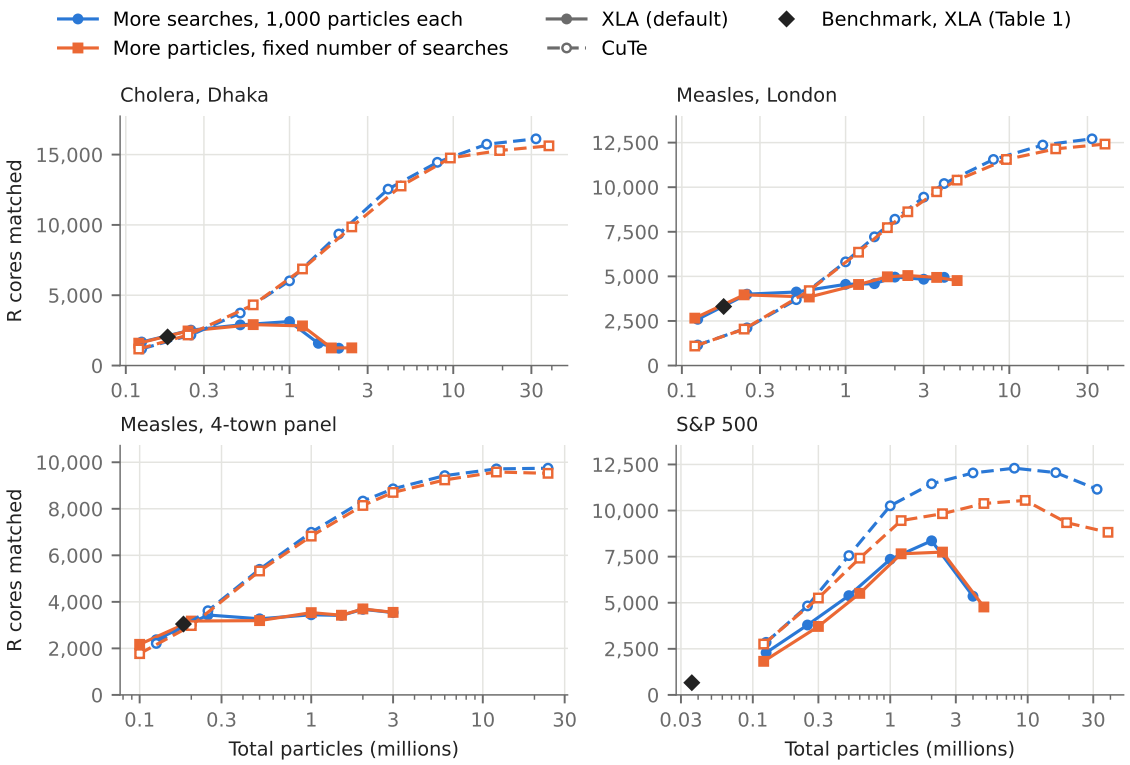


Figure 1: IF2 throughput on one GPU, as the number of CPU cores running R that the GPU matches, against the total number of particles in flight. Blue adds searches of 1,000 particles each; orange adds particles to each of 100 or 120 searches. Solid lines with filled markers use the default XLA backend of Pypomp 1.4.0, and dashed lines with hollow markers use its CuTe backend. Each configuration ran 3 or 10 IF2 iterations, projected to the benchmark's number. The x-axes use a log scale.

Table 4: Time in milliseconds for 1,000,000 draws from `pypomp.random` in Pypomp (version not recorded) and `jax.random`, on one GPU and on one CPU core, with parameters spread across the values where the samplers switch algorithms. Ratio is the `jax.random` time divided by the `pypomp.random` time. `jax.random` has no negative binomial sampler. Max W_1/σ is the largest Wasserstein distance between `pypomp.random` draws and draws from `jax.random` (or `scipy.stats` for the negative binomial) over a grid of parameter values, relative to the standard deviation. **Pypomp version not recorded:** for samplers/gpu, samplers/cpu, samplers/grids.

	GPU			CPU			
Distribution	`pypomp`	`jax`	Ratio	`pypomp`	`jax`	Ratio	Max W_1/σ
Poisson	0.04	1.01	25×	28.46	896	31×	0.010
Binomial	0.02	1.07	45×	16.50	928	56×	0.010
Gamma	0.03	0.77	23×	102	2,734	27×	0.011
Negative binomial	0.07	—	—	162	—	—	0.009

averaged over 20 calls. The approximation error is small: over a grid of parameter values, the Wasserstein distance between 1,000,000 draws from each `pypomp.random` sampler and from a reference sampler is at most 1.1% of the reference standard deviation. In the London measles benchmark, the `jax.random` samplers made IF2 45 times slower and particle filtering 22 times slower on the GPU.

Appendix E. England and Wales measles fit

We ran 36 MPIF searches, each with 10,000 particles for 200 iterations, and evaluated each endpoint with 36 particle filter replicates of 5,000 particles. We then combined the best estimate for each town across searches and evaluated the result with another 36 replicates. Table 5 gives the time for each stage on one GPU with Pypomp 1.4.0 and its CuTe backend.

For comparison, we estimated how long the same computation would take with panelPomp on a 36-core CPU node. We measured panelPomp's cost per particle, per IF2 iteration, and per town on the 4-town panel in Table 2, which covers the same period with the same daily time step, and scaled it linearly to the full computation. We assume panelPomp fits the towns jointly with MPIF, as Pypomp did, so each search runs on one core and does not parallelize across units – this would be necessary in the general case when using shared parameters. By this estimate, R would need about 4.8 months, 226 times longer than the GPU with the CuTe backend, and about 14.3 years on a single core, so the GPU delivered the throughput of about 8,139 CPU cores running R. The wall clock time can be reduced further by spreading the searches over multiple GPUs.

Table 5: Wall-clock time for fitting the measles model to all towns in England and Wales. Pypomp times were measured on one GPU, with Pypomp 1.4.0 and its CuTe backend. R panelPomp times are estimates for a 36-core CPU node, extrapolated from the 4-town panel in Table 2.

Stage	Pypomp (1 GPU) CuTe	R `panelPomp` 36 cores (estimated)
IF2	14.6 hours	136.0 days
PF, all search endpoints	50 minutes	9.3 days
PF, combined estimate	3 minutes	6.2 hours
Total	15.4 hours	145.5 days